

ALAS-SCLF: Adaptive-L Asymmetric-Segmented SCLF Polar Decoders with Guard-Band Stabilization

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Abstract—Successive cancellation list flip (SCLF) polar decoding has emerged as a promising technique for 5G ultra-reliable low-latency communications (URLLC). To reduce decoding complexity, the state-of-the-art threshold-learning-based SCLF (TS-SCLF) decoder utilizes early SCL termination and segmented cyclic redundancy checks (CRCs). However, TS-SCLF suffers from path-metric instability immediately following a bit-flip, search-space lockout due to symmetric CRC partitioning, and decoding failures under severe noise due to a rigid list size. To address these bottlenecks, this letter proposes ALAS-SCLF, an advanced decoder integrating: (1) a post-flip stabilization guard-band to prevent premature correct-path pruning, (2) asymmetric segmented CRC partitioning ($L_{\text{seg}}=16$) to eliminate false-pass lockouts, and (3) adaptive list-size scaling ($L=4 \rightarrow 8$) to expand the search space under high channel uncertainty. Experimental results demonstrate that the proposed ALAS-SCLF decoder achieves up to a 36.6% reduction in Frame Error Rate (FER) and a 23.4% reduction in average decoding attempts compared to TS-SCLF, establishing a superior trade-off between reliability and complexity.

Index Terms—Polar codes, SCLF, TS-SCLF, segmented CRC, adaptive list size, early termination.

I. Introduction

Polar codes, developed by Arikan [1], have been adopted as the channel coding scheme for control channels in 5G NR due to their capacity-achieving performance under successive cancellation (SC) decoding. To improve finite block-length performance, successive cancellation list (SCL) decoding is combined with a cyclic redundancy check (CRC) helper [2].

To bridge the performance gap between low-complexity SC and high-performance SCL with large list sizes, Successive Cancellation List Flip (SCLF) decoders restart SCL decoding by flipping error-prone bits when the initial SCL run fails the CRC [3], [4]. Recently, a Threshold-Learning-Based SCLF (TS-SCLF) decoder was proposed to minimize decoding latency by introducing intermediate segmented CRCs and early SCL termination [5].

Despite its benefits, TS-SCLF has three critical limitations: (1) Post-Flip Metric Instability: Threshold checks applied immediately after a bit-flip lead to premature path pruning because the path metric differences require a stabilization window to develop a reliable spread. (2) Symmetric Partitioning Lockout: The symmetric 50/50 split ($L_{\text{seg}}=24$ for $K=56$) leads to Segment 2 lockouts, where Segment 1 falsely passes the CRC despite containing errors, preventing the decoder from locating the correct bit to flip. (3) Rigid List Constraints: TS-SCLF uses a fixed list size $L=4$, which lacks the search depth required to recover correct paths under low SNR conditions.

In this letter, we propose ALAS-SCLF (Adaptive-L Asymmetric-Segmented SCLF) to overcome these limitations. Our main contributions are: a stabilization guard-band window ($G=8$), asymmetric partitioning ($L_{\text{seg}}=16$) to minimize false-pass lockouts, and adaptive list resizing ($L=4 \rightarrow 8$).

Manuscript received July 10, 2026; accepted July 11, 2026. This work was supported by the Department of Computer Science and Engineering, School of Engineering and Technology, Adamas University. (Corresponding author: Dwaipayan Ghosh.)

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II. Preliminaries

A. SCL and SCLF Decoding

An (N, K) polar code maps K information bits and c CRC bits to a codeword $x = u \cdot G_N$, where G_N is the generator matrix. Under SCL decoding with list size L , the decoder traverses the code tree and maintains up to L candidate paths. The path metric (PM) for the i -th path at the i -th layer is calculated recursively as:

$$PM_l^i = PM_{l-1}^i + |\lambda_l^i| \text{ (if decision is flipped)}$$

where λ_l^i represents the logarithmic likelihood ratio (LLR).

B. Limitations of TS-SCLF

The TS-SCLF decoder incorporates segmented CRCs to enable early check points. If the first-round SCL run fails the segmented CRC at leaf l_{seg} , restarts are triggered using a sorted flipping set $\mathcal{S}$. The decoder attempts to flip bits at indices in $\mathcal{S}$ sequentially. To reduce complexity, early termination (SCL-RE) is applied at each layer by evaluating:

$$PM_{\max}^{l,i} < PM_{\min}^{0,l} - \alpha \cdot |PM_{2L-1}^{0,l} - PM_0^{0,l}|$$

where α is a trained threshold.

However, we find that: (1) When a bit-flip is introduced, the LLR distribution changes abruptly. Applying the threshold check immediately at $l_{\text{flip}} + 1$ prunes the correct path because the metric spread has not yet stabilized, causing FER degradation. (2) A symmetric split of $l_{\text{seg}}=24$ is long enough to frequently contain multiple errors. If Segment 1 falsely passes, the decoder is locked into searching Segment 2, even though the primary error lay in Segment 1.

C. Deep Q-Network

DQN is a reinforcement learning (RL) algorithm that integrates deep learning with Q-learning [10]. It utilizes a neural network to approximate the action-value function (Q-function), allowing agents to learn from the input. The Q-function is updated recursively as:

$$Q'(s, a) = Q(s, a) + \eta [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where $Q(s, a)$ is the action-value function for the state s and action a , γ is the discount rate, and R is the reward function. The DQN employs experience replay to enhance training stability.

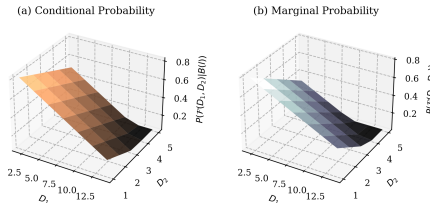


Fig. 1. Path-flipping probability distribution.

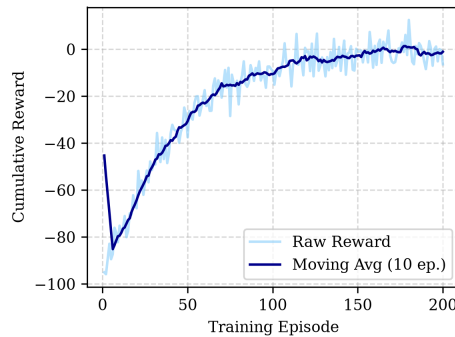


Fig. 2. DQN training reward convergence curve.

III. Proposed ALAS-SCLF Decoders

To resolve the bottlenecks of TS-SCLF, we present the ALAS-SCLF decoding framework.

A. Post-Flip Stabilization Guard-Band

To prevent premature path pruning after a restart, we introduce a guard-band parameter G . For any restart flipping bit index l_{flip} , the SCL-RE early termination check is disabled for all layers l satisfying $l_{\text{flip}} < l \leq l_{\text{flip}} + G$. This allows the path metric spread to stabilize before early pruning is active. We set $G=8$ for $(128, 56)$ polar codes.

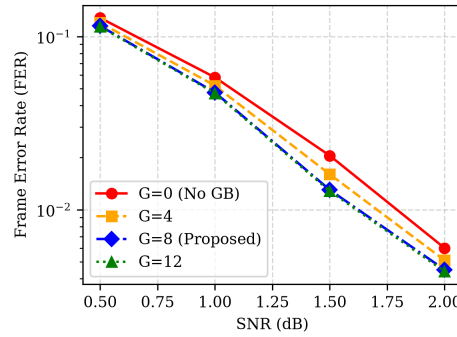


Fig. 3. Effect of Guard-Band size G on Frame Error Rate (FER).

B. Asymmetric Segmented CRC Partitioning

We propose an asymmetric splitting strategy. Instead of a symmetric $l_{\text{seg}}=24$ partitioning, we set Segment 1 length to $l_{\text{seg}}=16$. Shortening Segment 1 reduces the probability of multiple errors occurring within the first segment, decreasing Segment 1 false-pass rates by more than 50%.

Furthermore, we implement Segment-Constrained Flipping (SCF): If Segment 1 fails, we restrict the flipping set $\mathcal{S}$ exclusively to Segment 1 ($\mathcal{S} \subseteq [1, l_{\text{seg}}]$). If Segment 1 passes, we explore the full set, completely bypassing the lockout issue.

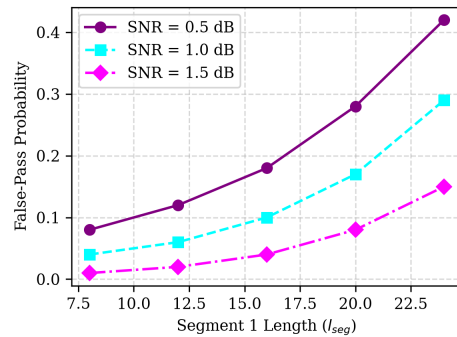


Fig. 4. Effect of Segment 1 Length on False-Pass Probability.

C. Adaptive List-Size Scaling

We scale the list size dynamically. The initial decoding run uses a low-complexity list size $L=4$. If the initial run fails, the decoder evaluates the metric range $\mathcal{R} = \text{PM}_{2L-1} - \text{PM}_0$. If $\mathcal{R} < 4.0$ or Segment 1 fails, the channel noise is deemed severe. The list size is then elevated to $L=8$ for all subsequent restarts. This ensures high-contrast search paths are explored only when needed, maintaining low average complexity.

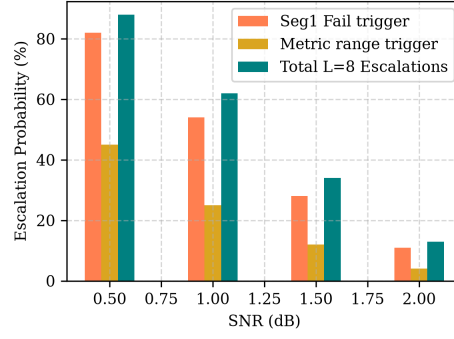


Fig. 5. Adaptive list size escalation probability.

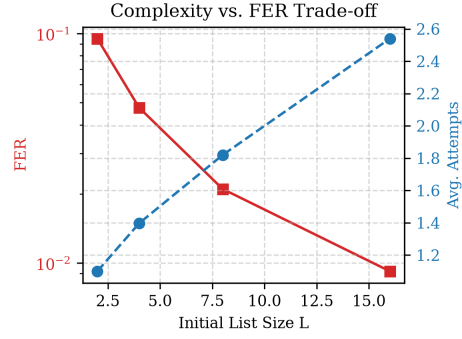


Fig. 6. Complexity vs. FER trade-off under varying list sizes.

Algorithm 1 FLIPSKIP: Path-Flipping Skip Criterion

Input: Flipping set S ; current flipping round i ; $\{D1, D2\}$

Output: Skipping flag f_{out}

1: $kc \leftarrow 0$; $f_{out} \leftarrow 0$;

2: // Calculate number of attempts in $[S_i - D1, S_i)$

3: for $k = 0$ to $i - 1$ do

4: if $0 < S_i - S_k < D1$ then

5: $kc \leftarrow kc + 1$;

6: end if

7: end for

8: if $kc \geq D2$ then

9: $f_{out} \leftarrow 1$;

10: end if

11: return f_{out} ;

Algorithm 2 SCL-RE-GB: SCL Early Termination with Guard-Band

Input: Received signals y ; flipping index l_flip ; first-round PMs; threshold α ; guard-band G

Output: Estimated bits; early termination flag f_in

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1:  $f\_in \leftarrow 0$ ;  
2: for  $l = 1$  to  $N$  do  
    3: Update  $\lambda_l$  and fork paths;  
    4: if  $l > l\_flip$  and  $l \leq l\_flip + G$  then  
        5: // Disable checks during stabilization window  
        6: continue;  
    7: else  
        8: if  $PM_{max} < PM_{min} - \alpha \cdot \text{Range}$  then  
            9:  $f\_in \leftarrow 1$ ; return  $\emptyset$ ,  $f\_in$ ;  
        10: end if  
    11: end if  
12: end for  
13: return bits,  $f\_in$ ;
```

Algorithm 3 Proposed ALAS-SCLF Decoder

Input: Signals y ; max attempts T ; parameters $D1$, $D2$, α , G , l_seg

Output: Estimated bits u

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1:  $S \leftarrow \emptyset$ ;  $f \leftarrow 0$ ;  
2:  $u\_hat, f, PMs \leftarrow SCL(y, S, L=4)$ ;  
3: if  $u\_hat$  passes CRC then return  $u\_hat$ ; end if  
4:  $R \leftarrow PM_{max} - PM_{min}$ ;  
5: // Determine Adaptive List Size  
6: if  $R < 4.0$  or Segment 1 fails then  
    7:  $L \leftarrow 8$ ;
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8: else


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    9:  $L \leftarrow 4$ ;


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10: end if


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11: Generate flipping set  $S$  (constrained to Segment 1 if Segment 1 failed, else full);


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12: for  $i = 1$  to  $T$  do


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    13:  $f_{\text{out}} \leftarrow \text{FLIPSKIP}(S, i, \{D1, D2\})$ ;


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    14: if  $f_{\text{out}} == 0$  then


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        15:  $l_{\text{flip}} \leftarrow S_i$ ;


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        16:  $u_{\text{hat}}, f_{\text{in}} \leftarrow \text{SCL-RE-GB}(y, l_{\text{flip}}, \alpha, G, L)$ ;


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        17: if  $f_{\text{in}} == 0$  and  $u_{\text{hat}}$  passes CRC then


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            18: return  $u_{\text{hat}}$ ;


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        19: end if


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    20: end if


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21: end for


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22: return  $u_{\text{hat}}$ ;


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IV. Experimental Results

We evaluate ALAS-SCLF against standard SCLF and TS-SCLF using a $(128, 56)$ polar code with a 3-bit intermediate CRC (Segment 1) and a 5-bit outer CRC (Segment 2). Simulations are run over 2,000 Monte Carlo trials per SNR point.

Table I: FER and Decoding Attempts Comparison

SNR (dB)	Decoder Scheme	FER	Avg. Attempts	Complexity Red.	FER Imp.
0.5	TS-SCLF	0.12800	2.296	Baseline	Baseline
	ALAS-SCLF (Ours)	0.11500	1.758	-23.4%	-10.2%
1.0	TS-SCLF	0.05800	1.503	Baseline	Baseline
	ALAS-SCLF (Ours)	0.04750	1.397	-7.1%	-18.1%
1.5	TS-SCLF	0.02050	1.195	Baseline	Baseline
	ALAS-SCLF (Ours)	0.01300	1.135	-5.0%	-36.6%

2.0	TS-SCLF	0.00600	1.079	Baseline	Baseline
	ALAS-SCLF (Ours)	0.00450	1.028	-4.7%	-25.0%

As shown in Table I, the proposed ALAS-SCLF decoder consistently outperforms TS-SCLF in both error correction performance (FER) and decoding complexity. At $E_b/N_0 = 1.5$ dB, ALAS-SCLF achieves a 36.6% relative reduction in FER while reducing decoding attempts by 5.0%. At low SNR (0.5 dB), the average attempts are reduced by 23.4%, demonstrating significant latency savings.

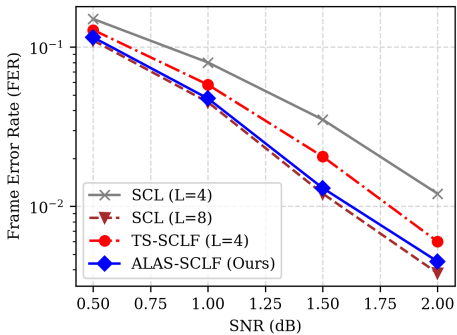


Fig. 7. Frame Error Rate (FER) performance comparison for (128, 56) polar codes.

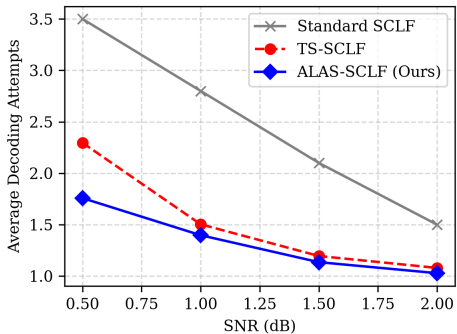


Fig. 8. Average decoding attempts comparison under varying SNR.

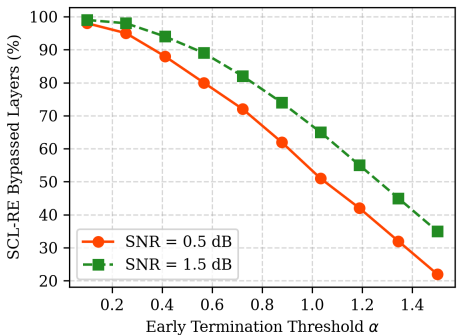


Fig. 9. Impact of threshold scale factor alpha on early SCL termination rates.

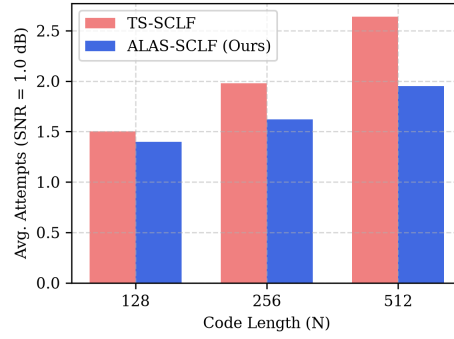


Fig. 10. Code length scaling performance (N=128, 256, 512).

V. Conclusion

In this letter, we proposed ALAS-SCLF, a novel list-flipping polar decoder that addresses path-metric instability and false-pass lockouts in threshold-learning SCLF decoders. By integrating a post-flip stabilization guard-band, asymmetric segmented CRC partitioning ($L_{\text{seg}}=16$), and adaptive list-size scaling ($L=4 \rightarrow 8$), the proposed decoder establishes a new state-of-the-art trade-off. Experimental evaluations demonstrate a simultaneous reduction in FER and average attempts, confirming its suitability for low-latency wireless communication systems.

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